Ling 285
Human Language and Computation
Lecture 1

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Speech and Language

- You learn to move your speech organs in extremely complex way with neither instructions, nor hardly any visual feedback.
- By 2-4 you can speak with a wide vocabulary and near-adult language, with hardly any explicit instruction.
- You may have also memorized the visual shape of the alphabet, but reading and writing whole wordscomes at 6-7, and requires a great deal of instruction.
- Articulatory Phonetics is the field that studies the movements of different vocal tract structures.
- And the last two year Voice Assistants and other Speech Tech have come to emulate the speech ability in ways none of us thought possible.
- Ling 285 is about Human and AI speech

Now think of the air between the tongue and the rest of the vocal tract. And ignore the fact that it is curved. This is how air vibrates in your vocal tract, schematically. That's the subject matter of Acoustic Phonetics.



AI: Ughhhhhh… Can’t we just ignore AI?

- Given all we know about the psychological, economic, social, and environmental harm, can we just ignore AI– just pretend it doesn’t exist?
- Given the existential harm that many scientists warn about, can we just pretend it doesn’t exist?
- It exists, with many harms, and some be[illegible]notherapy, and yes, environmental engineering)
- One of the biggest dangers: We have no idea how it works. ***We know everything, and Understand nothing***.
- And the people who *know* how it operates are very very few.
- In my opinion, AI literacy is necessary for scientists who use it, politicians who need to control it, lawyers who litigate on it, etc.
- I teach this course, because I believe that someone in a class like this, a creatve thinker can help us understand these systems enough to benefit from them, while avoiding their negative consequences

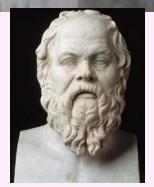


Goals of Ling 285

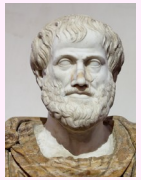
- Specific:
  - To appreciate the human cognitive skill of speech
  - To understand some basic principles of physics through speech
  - To understand how modern AI systems model human cognition
  - To sharpen your methodical reasoning via elementary programming
- General:
  - How to think creatively in a rich interdisciplinary way, as this course will involve physics, psychology, linguistics, biology, economics, computer science, philosophy, neuroscience, and many other fields.
  - To deepen your understanding of the world we are all living in today
  - To sharpen your skillful use of modern AI systems

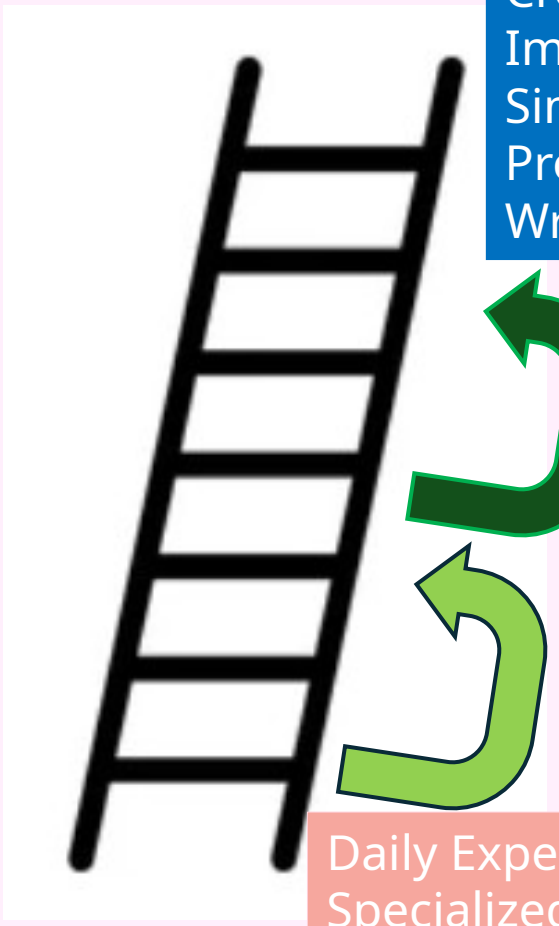
Diotima (440 BCE) to Job Interviews (2026…)











Creativity, Insight, Thought, Focus, Attention, Imaginative, Vision, different can be same, Similar can be different, Problem-Posing, Problem-Solving, Critical Reading and Creative Writing

Question-Answer

Question-Answer

Question-Answer

Question-Answer

Daily Experience, Habit, Memorization Specialized skills, Repetition and Replication, Similar is Same, and Slight difference is different, textbooks

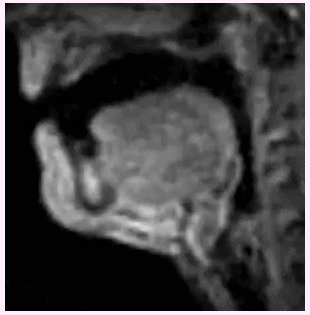
For 2500 years, people could fake it! They could do well without ascending the ladder… Sorry… No more.
Not in 2026. For instance, In interviews
for jobs and grad school, all anyone is interested anymore is whether you exhibit properies of the Top of
Diotima's ladder!

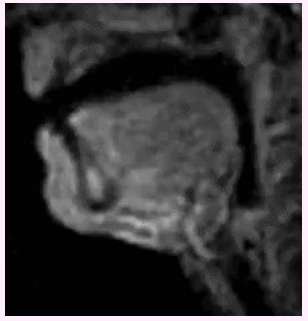
Speech: Let’s learn by doing

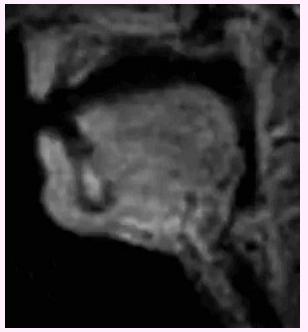
- fffffffffffffvvvvvvvvvvvv (say the f in fat, then v in vat)
- thththththththththththth(say the th in thing, then th in that)
- sssssssssszzzzzzzzzzzz(say the s in sing, then z in zing)
- shshshshshjjjjjjjjjjjjjjjjjj (say the sh in mesh, then j in beige)
- What’s different between the first and second elements of each pair?
- Voice: Whether your vocal folds vibrate or not.
- What’s the difference between the elements on different lines?
- Place of Articulation

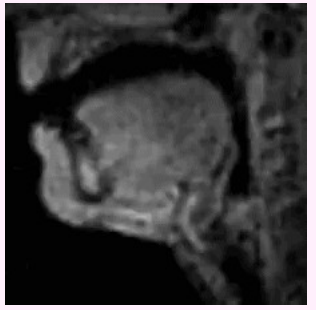
Fricatives

	Labial	Dental	Alveolar	Postalveolar
Voiceless	f	th	s	sh
Voiced	v	th	z	j



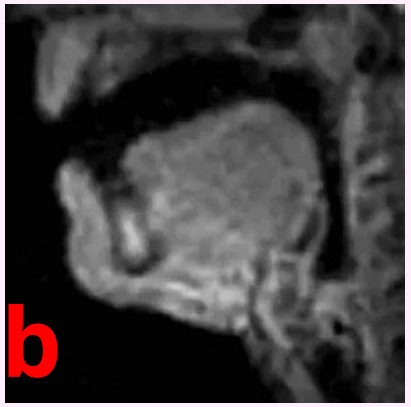






Stops

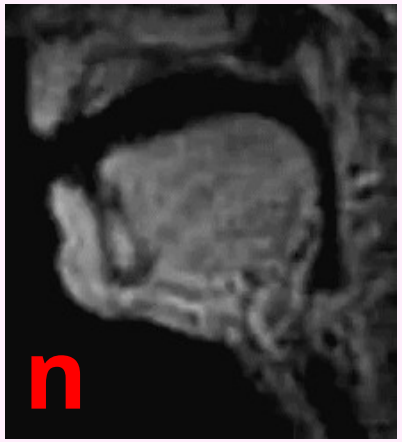








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